

• Optimal Transport (Monge Formulation)

source distribution ρ , target dist. μ

find the map T s.t.

$$\min C(T) = \int \underbrace{c(x, T(x))}_{\text{quadratic cost}} \rho(x) dx \quad \text{s.t. } T\#\rho = \mu$$

Eg. quadratic cost $c(x, y) = \frac{1}{2} \|x - y\|^2$

Lagrange
 $\Rightarrow$

Let $z := T(x, t)$ (map/flow at time t)

Dual

$$T(x, 0) = x$$

particle flow $\begin{cases} \dot{z} = -\nabla_z \left[\frac{\partial \tilde{L}_t}{\partial u} \right]_{u = \frac{\|z\|^2}{2}} = -\nabla_z (\rho_t(z) - \mu(z)) \\ z(t=0) = x \end{cases}$

(Gradient Flow)

at time t , $\{z_i\} \sim \rho_t$

• Data-Driven case : $\{x_i\}_{i=1}^N \sim \rho$, $\{y_j\}_{j=1}^M \sim \mu$

$$z_i = x_i$$

$$T(z_i, N) \sim \mu$$

large time step

$\frac{\partial}{\partial z_i}$

$$\Rightarrow \text{Gradient Descent : } z_i^{n+1} = z_i^n - \eta \cdot \nabla_z (\rho_t(z) - \mu(z)) \Big|_{z_i^n}$$

(Discretize flow)

learning rate

$$k(z_i^n, y_j) \Big|_{i=1, \dots, N, j=1, \dots, M}$$

$$\textcircled{1} \text{ KDE : } \nabla_z (\rho_t(z) - \mu(z)) \approx \frac{1}{N} \sum_{i=1}^N \nabla_z K_h(z, z_i^n) - \frac{1}{M} \sum_{j=1}^M \nabla_z K_h(z, y_j)$$

$$\{z_i\}_{i=1}^N \sim \rho_t$$

$$\{y_j\}_{j=1}^M \sim \mu$$

kernel matrix $k(z_i^n, z_j^n) \Big|_{i=1, \dots, N, j=1, \dots, N}$

$$\textcircled{2} \quad \rho_t(z) - \mu(z) = \int \delta(z-z') (\rho_t(z') - \mu(z')) dz' \quad \delta(t) = \begin{cases} 1, & t=0 \\ 0, & t \neq 0 \end{cases}$$

$$\approx \int \underbrace{k(z, z')}_{\text{"local window"}} (\rho_t(z') - \mu(z')) dz'$$

$$= \int k(z, z') \rho_t(z') dz' - \int k(z, z') \mu(z') dz'$$

Monte Carlo

$$\approx \frac{1}{N} \sum_i k(z, z_i^i) - \frac{1}{M} \sum_j k(z, y_j)$$

- KDE (Kernel Density Estimate)

- Non-parametric

- Parametric: $\mathcal{P} = \{P_\theta : \theta \in \Theta\}, \Theta \subseteq \mathbb{R}^p$ (finite dim)

- Eg. $N(\mu, \sigma^2), p=2$

$\uparrow \quad \uparrow$
 $\mu \quad \sigma^2$

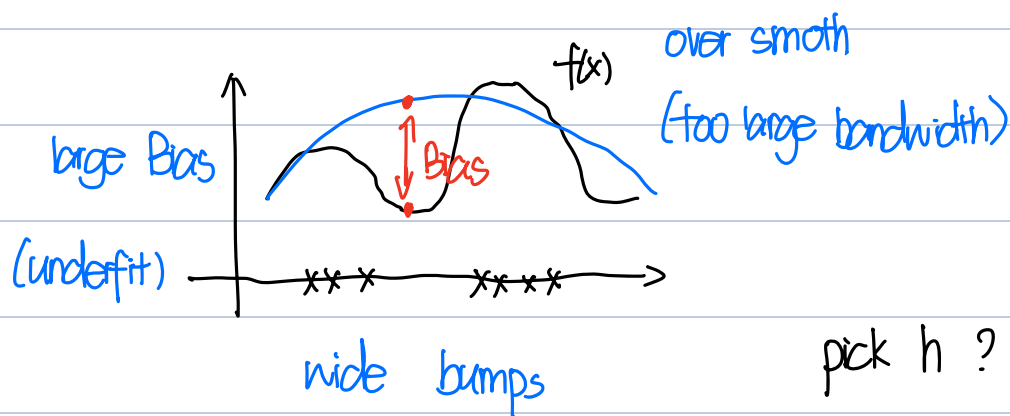
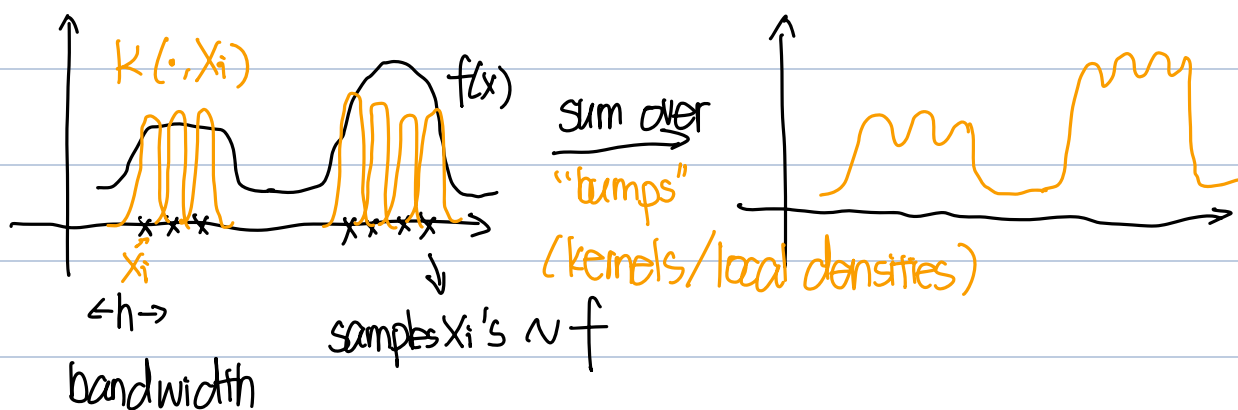
Now, $\mathcal{P} = \{\text{all cts distribution on } \mathbb{R}\}$ (1D)

$$= \{f: \mathbb{R} \rightarrow \mathbb{R} \mid f \geq 0, \int f = 1\}$$

Question: Can we estimate f (based on data/stats) ?

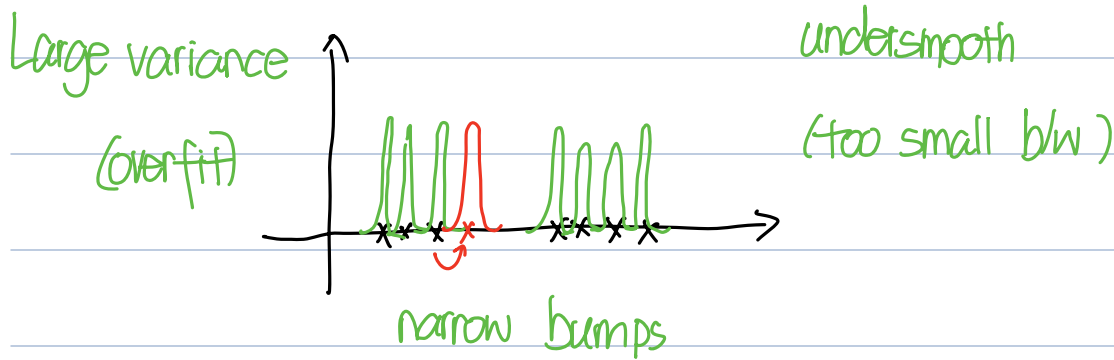
(Under smoothness assumption for f)

- KDE



pick h ? Eg. Rule of Thumb

(-Thm: Bias-Variance Trade off
 $\Rightarrow h \sim n^{-1/3}$)



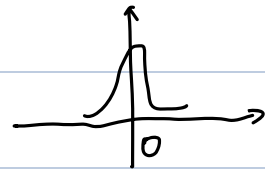
• A kernel is a func. $K: \mathbb{R} \rightarrow \mathbb{R}$

① $\int K(u) du = 1$

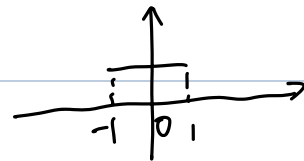
② $\int u K(u) du = 0$

$\Rightarrow K_h(\cdot, z) \sim N(z, h^2)$

-Eg. (1) $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$



(2) $K(u) = \frac{1}{2} \cdot \mathbb{1}_{[-1,1]}$



We will consider for $h > 0$, $z \in \mathbb{R}$
 $\hookrightarrow$ bandwidth

$\frac{1}{h} K\left(\frac{\bullet - z}{h}\right)$ rescaled version of K centered at z

$X_1, \dots, X_n \sim$ from unknown dist.

our estimator $\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right) = \frac{1}{n} \sum_{i=1}^n K_h(x, X_i)$

$\uparrow$ normalizing factor $\Rightarrow \int \hat{f}(x) dx = 1$